A WEB of Sharing Trust–Responsibility Toward a Trustless Future vol.2

Core Governance Paper with Embedded CSC Scoring
Second Edition — Version A

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Distribution Statement

This publication is released as a **Public Review Draft (Standards Exploration)**.

It may be used for:

- academic research and comparative analysis,
- policy development and regulatory consultation,
- standards exploration and working-draft discussion, and
- evaluation of approaches to AI governance that move beyond static, documentation-based compliance toward **engineered, runtime trust mechanisms**.
- Feedback is invited to support refinement prior to subsequent editions and related artifacts.

This document does **not** constitute regulatory guidance, certification, or legal advice. It is intended to support informed review, discussion, and refinement of governance architectures, assurance models, and scoring methodologies for AI systems.

Edition Note

The Second Edition expands upon the original publication by:

- Introducing machine-readable governance documentation (JSON artifacts)
- Embedding the Compliance Star Certificate (CSC) Scoring Rubric directly within the core manuscript
- Aligning terminology, structure, and normative language with ISO/IEC standards practices
- Reflects advances in Simulation AI-Powered Policies (SAPP), runtime governance, and alignment with emerging international AI standards.

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Abstract

As artificial intelligence systems scale in autonomy, capability, and societal reach, governance approaches based on static compliance, documentation, and post-hoc audits do not provide lifecycle assurance. Contemporary AI systems are adaptive, probabilistic, and increasingly composed of multiple models, tools, and services, introducing risks that cannot be adequately managed through point-in-time assessment alone.

This document defines a model-agnostic, execution-independent framework for **operational, evidence-based AI governance**, establishing how trust is **engineered, measured, and revoked** using runtime evidence across the AI system lifecycle. The framework formalizes three complementary mechanisms: **Simulation AI-Powered Policies (SAPP)** for pre-deployment and early-deployment risk discovery, a **Policy-on-Demand (PoD) run-time governance engine** for enforceable control execution, and a **telemetry-derived Compliance Star Certificate (CSC)** that provides continuous, revocable assurance.

CSC assurance **shall be derived exclusively from observed system behavior**, rather than declarative compliance statements or static documentation. Governance effectiveness is evaluated through non-compensatory scoring, mandatory qualification gates, and lifecycle-aware re-scoring, aligning with ISO/IEC risk-management principles without introducing architecture-specific dependencies.

This Version A document is intended to serve as a **top-level governance and assurance reference** for policymakers, standards bodies, researchers, and system designers. It is deliberately architecture-agnostic and model-provider-neutral, enabling adoption across diverse AI systems, including foundation models and agentic systems. A subsequent Version B, scheduled for February, will introduce an optional architecture-specific extension using the Q-Pathformer system as a reference implementation.

1. Introduction

Artificial intelligence systems are increasingly deployed as **long-lived, adaptive infrastructures** rather than static software artifacts. As a result, governance approaches that rely primarily on documentation, one-time assessments, or post-hoc audits do not provide meaningful lifecycle assurance. This document introduces an operational governance framework designed to support **continuous, evidence-based trust** in AI systems as they evolve over time.

1.1 The Limits of Static AI Compliance

Traditional AI governance practices have emphasized **static compliance mechanisms**, including policy declarations, model cards, risk registers, and periodic audits. While such mechanisms **may** provide baseline accountability at a point in time, they **do not provide lifecycle assurance** for systems that:

- are retrained, fine-tuned, or updated after deployment,
- incorporate third-party models, tools, or data sources,
- operate across heterogeneous environments and jurisdictions, or
- exhibit emergent behavior through interaction with users and other systems.

Static compliance frameworks **shall not be considered sufficient** to demonstrate ongoing conformity with safety, ethical, or regulatory expectations. In particular, they fail to capture **runtime behavior**, **control degradation**, and **contextual misuse** that may arise after initial approval. These gaps create a false sense of assurance and limit the ability of regulators, operators, and stakeholders to detect and respond to material risk.

This document addresses these lifecycle assurance gaps by treating governance as an **operational system**, continuously evaluated through observable evidence rather than inferred intent.

1.2 From Trust Declarations to Trust Engineering

Many existing assurance approaches rely on **attestation**, whereby organizations declare that specified controls, principles, or processes are in place. Attestations **assert intent or design**, but they **do not verify behavior**.

This document distinguishes clearly between:

- **Attestation-based assurance**, which is document-centric and static; and
- **Telemetry-based assurance**, which is evidence-centric and continuous.

Trust engineering **shall** be grounded in telemetry that reflects how an AI system actually behaves under simulation and during runtime operation. Within this framework:

- governance claims **shall be supported by measurable evidence**;
- assurance **shall be revocable** when evidence degrades; and
- trust **shall be earned continuously**, not granted indefinitely.

Simulation AI-Powered Policies (SAPP), the Access Policy-on-Demand (PoD) engine, and the Compliance Star Certificate (CSC) together implement this shift by embedding governance into **system execution paths**, rather than external reporting artifacts.

1.3 Scope and Applicability Across AI & ML Systems

This document specifies a **governance and assurance layer that operates above AI system architectures**, focusing on execution-time controls, telemetry, and evidence-based trust rather than model internals or training methodologies.

The framework **shall apply** across a broad range of AI and ML system types, including:

- classical machine learning pipelines;
- computer vision systems;
- deep learning architectures;
- foundation models and large language models; and
- agentic or tool-using AI systems.

The framework is **architecture-agnostic and model-provider-neutral**. It does not prescribe specific algorithms, training techniques, performance benchmarks, or optimization strategies. Nor does it replace sector-specific regulatory requirements or constitute legal advice.

Instead, this document defines:

- how governance controls are executed at runtime;
- how evidence is generated, aggregated, and verified;
- how trust is quantified through non-compensatory scoring; and
- how assurance degrades or is revoked when evidence no longer supports prior trust claims.

Informative Note — Architecture-Specific Extensions

This Version A document intentionally excludes architecture-specific execution models, cryptographic provenance mechanisms, and model-internal telemetry schemas. A subsequent Version B, currently planned for February (subject to review outcomes), will introduce an **optional architecture-specific extension** using the **Q-Pathformer system** as a reference implementation. That extension will demonstrate how the Version A framework may be instantiated at execution depth, including additional telemetry dimensions and UGCCMT-based cryptographic provenance mechanisms.

The inclusion of Q-Pathformer in Version B **does not modify or constrain** the applicability of this Version A framework, which remains valid independently of any specific AI architecture.

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2. Conceptual Foundations

This section defines the conceptual foundations of the framework, establishing trust, transparency, and responsibility as **operational system properties**. It further introduces a **human-centric system maturity progression**, first articulated in Volume 1, and refined in this second volume to support telemetry-driven governance and lifecycle assurance.

2.1 Trust, Transparency, and Responsibility in AI Systems

For the purposes of this document, **trust** is defined as justified confidence that an AI system behaves within acceptable bounds under defined conditions and over time**.** Trust is not an intrinsic attribute of a model; it is an emergent property resulting from governance controls, system behavior, human oversight, and contextual constraints.

Transparency enables trust by making system behavior observable, explainable, and auditable. Transparency **shall not be limited** to static documentation or disclosures; it **shall include runtime visibility** into decision pathways, control enforcement, and system state.

Responsibility establishes accountability for outcomes. Responsibility **shall be assignable, traceable, and enforceable** across organizational, technical, and operational boundaries, including post-deployment modification, reuse, and integration with external systems.

Together, trust, transparency, and responsibility form the **minimum conceptual triad** required for sustainable AI governance.

2.2 Human-Centric Governance as a System Property

Human-centric governance treats human judgment not as an external safeguard, but as an **integrated system control**. Human oversight **shall be designed into** workflows, escalation logic, and assurance mechanisms rather than applied retrospectively.

Within this framework:

- human intervention **shall be available** where risk exceeds predefined thresholds;
- decisions **shall be contestable** through traceable evidence; and
- escalation pathways **shall be explicit, proportionate, and auditable**.

Human-centric governance therefore operates as a **first-class system property**, comparable to safety, security, or reliability controls, and not as an informal organizational practice.

2.3 System Maturity Progression (UIO-Defined Technologies)

Volume 1 introduced a **human-centric maturity model** for emerging technologies, describing how systems evolve from novelty to operational stability. This second volume refines that model by explicitly aligning each maturity stage with **risk characteristics, governance needs, and assurance expectations**.

The maturity progression consists of three stages: **Under-defined**, **Informally defined**, and **Over-defined**.

2.3.1 Under-Defined Systems (Exploratory / High Uncertainty)

An **under-defined** system is one that is novel or emerging and may appear useful, compelling, or innovative, but whose behavior, boundaries, and impacts are not yet well understood.

Such systems:

- may sound, display, or express usefulness and interest;
- often lack stable operating assumptions;
- may introduce confusion, misuse, or unanticipated risk; and
- exhibit high uncertainty regarding downstream impact.

From a governance perspective, under-defined systems correspond to **high uncertainty and unquantified risk exposure**. Assurance at this stage **shall focus on exploration, simulation, and risk discovery**, rather than certification or deployment confidence. In this framework, Simulation AI-Powered Policies (SAPP) are intended to operate primarily at this stage to surface latent risks before real-world impact occurs.

2.3.2 Informally Defined Systems (Operationalizing / Managed Risk)

An **informally defined** system is one for which there is growing enterprise or consumer uptake and a shared understanding of expected behavior, even if full formalization has not yet occurred.

Such systems:

- follow emerging procedures and protocols;
- reflect a developing human-centric approach to use and governance;
- have partially defined controls and responsibilities; and
- exhibit identifiable but not fully stabilized risk profiles.

At this stage, risk is **identified and partially treated** but remains context dependent. Governance mechanisms **should support continuous evaluation**, operational telemetry, and adaptive controls. Informally defined systems are suitable candidates for **run-time governance and provisional assurance**, but not yet for static or permanent certification.

2.3.3 Over-Defined Systems (Stabilized / Controlled Risk)

An **over-defined** system is one where expectations, boundaries, and behaviors are well understood by stakeholders.

Such systems:

- have clear operating assumptions and constraints;
- provide a reliable overview of expected behavior;
- support predictable outcomes within defined bounds; and
- enable confident planning regarding future evolution.

From an ISO/IEC risk-management perspective, over-defined systems correspond to controlled and monitored risk, provided that governance evidence remains current. These systems are suitable for

comparative assurance, certification, and benchmarking, subject to continuous telemetry-based validation.

2.4 Progression from Volume 1 to Volume 2

Volume 1 introduced the conceptual maturity language of **under-defined, informally defined, and over-defined** technologies as a human-centric lens for understanding innovation.

Volume 2 extends this model by:

- explicitly linking each maturity stage to **governance mechanisms**;
- introducing **telemetry-driven assurance** as a requirement for progression; and
- embedding maturity assessment into the **Access PoD, SAPP, and CSC frameworks**.

This progression reflects a shift from **conceptual understanding** to **operational governance**, without discarding the human-centric foundations established in the earlier work.

2.5 Alignment with International AI Standards

The conceptual model defined in this section is designed to align with international standards while remaining implementation-agnostic:

- **ISO/IEC 42001**: by treating governance as a management system with continuous improvement;
- **ISO/IEC 23894**: by recognizing risk as dynamic and context-dependent; and
- **ISO/IEC 22989**: by maintaining consistent terminology and system definitions.

This document does not replace these standards. It provides an **operational interpretation** of their principles, suitable for evaluation through evidence rather than documentation alone.

Normative Summary (Section 2)

- AI systems **shall be assessed relative to their maturity stage**.
- Governance **shall scale with uncertainty and impact**.
- Progression between maturity stages **shall be supported by evidence**, not assumption.
- Human-centric considerations **shall remain integral** at all stages of system evolution.

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3. Simulation AI-Powered Policies (SAPP)

Simulation AI-Powered Policies (SAPP) provide the **pre-deployment and early-deployment governance mechanism** within the Access PoD framework. SAPP operationalizes governance by executing policies as code within controlled simulation environments, enabling risk discovery, control validation, and evidence generation **prior to and during system deployment**.

SAPP is designed to support **different governance objectives at different system maturity stages**, as defined in Section 2.3. It therefore functions as a **maturity-aware governance layer**, rather than a one-time assessment tool.

3.1 Purpose and Design Principles

The primary purpose of SAPP is to **surface, evaluate, and constrain risk before real-world impact occurs**. SAPP **shall** enable organizations to test governance assumptions under realistic and adversarial conditions without exposing users, markets, or infrastructure to unvalidated behavior.

SAPP is governed by the following design principles:

- **Simulation-first assurance**: risk shall be explored through simulation before reliance on runtime controls.
- **Human-centric evaluation**: simulation outcomes shall be interpretable and reviewable by human stakeholders.
- **Proportional governance**: the depth of simulation shall scale with uncertainty and potential impact.
- **Evidence generation**: simulation outputs shall produce machine-readable telemetry suitable for downstream assurance.

3.2 Under-Defined Systems → Simulation-First Governance

For **under-defined systems** (Section 2.3.1), uncertainty regarding behavior, impact, and misuse is high. At this stage, governance **shall prioritize simulation over deployment**.

For under-defined systems, SAPP **shall**:

- execute policies against synthetic and anonymized real-world scenarios;
- explore misuse, edge cases, and adversarial interaction patterns;
- identify latent risks not apparent from design documentation; and
- generate evidence to inform whether deployment should proceed at all.

At this stage:

- runtime certification **shall not be issued**;

- CSC scoring **shall not be applied** beyond exploratory analysis; and
- governance decisions **shall be human-reviewed**.

Simulation-first governance ensures that novelty and apparent usefulness do not bypass necessary risk discovery.

3.3 Informally Defined Systems → Runtime Monitoring and Adaptive Controls

For **informally defined systems** (Section 2.3.2), expected behavior and usage patterns are partially understood, but risk remains context-dependent. At this stage, governance **shall transition from simulation-only to combined simulation and runtime oversight**.

For informally defined systems, SAPP **should**:

- continue targeted simulations for newly observed behaviors;
- validate runtime controls against evolving usage patterns; and
- feed telemetry into the Access PoD engine for continuous monitoring.

Runtime governance at this stage **shall include**:

- adaptive control thresholds;
- human-in-the-loop review for elevated-risk events; and
- provisional assurance signals based on observed behavior.

CSC scoring **may be generated** at this stage but **shall be explicitly marked as provisional** and subject to frequent re-evaluation.

3.4 Over-Defined Systems → Continuous Certification via CSC

For **over-defined systems** (Section 2.3.3), operating boundaries and expected behaviors are well understood. At this stage, governance emphasis **shall shift from exploratory simulation to continuous assurance**.

For over-defined systems:

- SAPP **shall** be used to validate significant system changes, retraining events, or context shifts;
- runtime telemetry **shall** be the primary evidence source; and
- CSC certification **shall** be continuously calculated and maintained.

Certification at this stage **shall not be static**. CSC scores **shall be recalculated** as telemetry changes, and certification **shall be downgraded, suspended, or revoked** where evidence no longer supports prior trust levels.

SAPP therefore remains active throughout the lifecycle, but its role transitions from **risk discovery** to **assurance reinforcement**.

3.5 Experiment–Experience–Deployment Model

SAPP evaluates governance effectiveness across three interdependent dimensions:

- **Experiment**: simulated policy execution under controlled conditions.
- **Experience**: human interaction, interpretability, and oversight effectiveness.
- **Deployment**: operational stability, control performance, and impact at scale.

Governance maturity **shall be assessed across all three dimensions**. Progression in one dimension **shall not compensate** for deficiencies in another.

3.6 Human-in-the-Loop Validation

Human oversight is a **mandatory control** within SAPP. For all maturity stages:

- simulation outcomes **shall be reviewable**;
- escalation triggers **shall be human-visible**; and
- final governance decisions **shall not be fully automated**.

Human-in-the-loop validation ensures that SAPP supports **accountable decision-making**, rather than opaque risk scoring.

Normative Summary (Section 3)

- Under-defined systems **shall use simulation-first governance**.
- Informally defined systems **should combine simulation with runtime monitoring**.
- Over-defined systems **shall maintain continuous certification via CSC**.
- SAPP **shall remain active across the entire system lifecycle**.

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4. The Access PoD Engine

The Access Policy-on-Demand (PoD) engine provides the **runtime governance, enforcement, and evidence-generation layer** of the framework. It operationalizes governance by executing policies as code and enforcing them at defined control points across AI system lifecycles.

Access PoD **shall function as a continuous control system**, rather than a static compliance mechanism, enabling real-time oversight, proportional intervention, and telemetry-based assurance.

4.1 Policy-as-Code and Runtime Enforcement

Within the Access PoD framework, governance policies **shall be implemented as executable policy-as-code artifacts**. These policies **shall be enforceable at runtime**, rather than relying on external procedures or post-hoc review.

Policy-as-code **shall**:

- define allowable and disallowed system behaviors;
- specify thresholds for intervention and escalation;
- encode sector-specific and context-specific constraints; and
- produce machine-readable outcomes for audit and assurance.

Runtime enforcement ensures that governance controls **operate concurrently with system execution**, enabling immediate response to deviations rather than delayed corrective action.

4.2 Internal and External Model Delegation

Modern AI systems frequently rely on **composed architectures**, combining first-party models with external foundation models, APIs, tools, and services. The Access PoD engine **shall explicitly govern delegation boundaries** between internal and external components.

Delegation controls **shall**:

- identify whether a model or service is internal or external;
- specify permitted scopes of interaction and data exchange;
- enforce policy constraints prior to and after delegation; and
- capture telemetry associated with delegated actions.

Responsibility **shall not be implicitly transferred** through delegation. Where external models are used, the Access PoD engine **shall maintain governance visibility and accountability** for outcomes within the defined scope of use.

4.3 Enforcement Points as Control Objectives

Access PoD **shall enforce policies at defined enforcement points**, each of which constitutes a **control objective** for governance and assurance purposes.

Enforcement points **shall include** where applicable:

- inference and decision gateways;
- tool invocation and action execution boundaries;
- data access, transformation, and transfer interfaces;
- human-in-the-loop review and escalation queues; and
- model update, retraining, and configuration change events.

Each enforcement point **shall be uniquely identifiable**, policy-addressable, and auditable. Failure to enforce policy at a defined enforcement point **shall be treated as a control failure** for assurance and scoring purposes.

4.4 Telemetry Capture and CSC Evidence Alignment

Telemetry captured by the Access PoD engine **shall form the primary evidence base** for CSC scoring and certification.

Telemetry **shall**:

- be structured and machine-readable;
- be attributable to specific enforcement points and policy controls;
- include timestamps, decision outcomes, and contextual metadata; and
- support traceability across system components and lifecycle events.

Telemetry capture **shall be continuous** and **shall not be selectively enabled** solely for certification events. Where telemetry is incomplete, degraded, or unavailable, CSC scoring **shall be constrained or suspended** accordingly.

This alignment ensures that CSC certification reflects **actual governance performance**, not inferred or simulated compliance.

4.5 The Six Operational Factors as Governance Domains

The Access PoD engine structures governance across six operational factors, each representing a **distinct governance domain**:

1. Industry 4.0 & Security
2. Security & New Best Practice
3. New Best Practice & Compliance
4. Compliance & Human-Centric Path
5. Human-Centric & Trust-Transparency
6. Trust-Transparency & Industry / Sector Alignment

Each operational factor **shall**:

- have associated policy controls;
- generate factor-specific telemetry;
- contribute independently to CSC scoring; and
- be evaluated without compensatory averaging.

This structure prevents isolated strengths from masking critical governance weaknesses.

4.6 Accountability Boundaries and Escalation

Access PoD **shall define explicit accountability boundaries** between automated controls, human operators, and organizational responsibility.

Escalation mechanisms **shall**:

- trigger when predefined thresholds are exceeded;
- route decisions to appropriate human reviewers;
- be logged and auditable; and
- support proportional intervention.

Automated systems **shall not** resolve high-impact governance decisions without human oversight. Accountability **shall remain traceable** across system components, organizational roles, and external dependencies.

Normative Summary (Section 4)

- Governance policies **shall be executable and enforced at runtime**.
- Delegation boundaries **shall be explicitly governed and auditable**.
- Enforcement points **shall function as control objectives**.
- Telemetry **shall directly support CSC evidence requirements**.
- Accountability **shall not be obscured by automation or delegation**.

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5. The Compliance Star Certificate (CSC)

The Compliance Star Certificate (CSC) provides a **continuous, telemetry-derived assurance mechanism** for AI systems governed under the Access PoD framework. CSC **shall not be interpreted as a static certification or compliance label**. Instead, it functions as a **dynamic trust signal** derived from observable system behavior over time.

CSC assurance is generated from evidence produced through Simulation AI-Powered Policies (SAPP) and runtime enforcement via the Access PoD engine. Certification **shall reflect current governance performance**, not historical approval.

5.1 From Compliance Artifacts to Continuous Assurance

Traditional compliance approaches rely on artifacts such as policies, attestations, and audit reports. These artifacts **describe intended controls**, but they do not verify that controls remain effective during operation.

CSC replaces artifact-based assurance with **continuous evidence-based assurance**. Within this framework:

- certification **shall be derived from telemetry**, not documentation alone;
- assurance **shall degrade** when evidence quality or control effectiveness degrades; and
- governance **shall be reassessed automatically** as system behavior evolves.

CSC **shall not be issued** in the absence of sufficient telemetry. Where evidence is incomplete or unavailable, CSC output **shall be constrained, suspended, or withheld**.

5.2 CSC Star Levels and ISO-Style Risk Tiers

CSC star levels represent **relative risk posture and governance maturity**, not quality, performance, or approval.

Star levels **shall be mapped explicitly** to ISO-aligned risk tiers, reflecting the extent to which identified risks are controlled, monitored, and treated.

CSC Level	Risk Tier Interpretation
★	Unacceptable risk
★★	High risk
★★★	Medium–High risk

CSC Level	Risk Tier Interpretation
★★★★	Medium risk
★★★★★	Low risk
★★★★★★	Minimal / Sector-Validated risk

This mapping ensures that CSC output is **interpretable within established risk-management frameworks**, including ISO/IEC 23894. Star levels **shall not** be interpreted as endorsements or suitability guarantees.

A system may achieve strong security scores but still receive a lower CSC rating if human-centric or transparency requirements are not met.

5.3 Certificate Lifecycle and Revocation Model

CSC certification **shall be continuous and lifecycle-aware**. A CSC **does not represent a point-in-time approval** and **shall not remain valid in the absence of supporting evidence**.

The CSC lifecycle **shall include**, at a minimum:

- issuance based on sufficient telemetry;
- continuous re-evaluation as new telemetry is generated;
- downgrade where evidence indicates increased risk or control degradation;
- suspension where evidence is incomplete or unreliable; and
- revocation where critical governance failures are identified.

Revocation **shall not be treated as punitive**. It is a **protective control**, indicating that trust claims are no longer supported by evidence. Re-issuance **shall require renewed evidence**, not administrative reinstatement.

5.4 Avoiding Certification-as-Status Ambiguity

CSC **shall not be treated as a status badge, marketing signal, or permanent designation**. Any use of CSC output **shall reflect its dynamic nature** and be accompanied by current risk interpretation.

Specifically:

- CSC output **shall not be cached or displayed without timestamp and validity context**;
- historical CSC levels **shall not substitute** for current assurance; and
- changes in CSC level **shall be considered material governance events**.

This ensures that CSC remains aligned with **runtime trust**, rather than devolving into symbolic certification.

Normative Summary (Section 5)

- CSC **shall function as continuous assurance**, not static certification.
- Star levels **shall represent risk posture**, not approval or quality.
- Certification **shall degrade or revoke automatically** when evidence no longer supports trust claims.
- CSC **shall not exist independently of telemetry**.

Certificates are continuously evaluated and may be downgraded or revoked as conditions change.

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6. CSC Scoring Rubric (Derived from SAPP Telemetry)

The Compliance Star Certificate (CSC) scoring rubric defines the **formal method by which governance performance is quantified** using telemetry generated through Simulation AI-Powered Policies (SAPP) and runtime enforcement via the Access PoD engine. The rubric **shall be evidence-driven, non-compensatory, and continuously updateable**, consistent with ISO/IEC risk management practices.

CSC scores **shall be computed**, not assigned, and **shall not exist independently of telemetry evidence**.

6.1 Telemetry Domains and Evidence Sources

CSC scoring **shall be derived exclusively from telemetry evidence** generated during simulation and runtime operation. Certification **shall not rely on declarative compliance statements, documentation alone, or self-attestation**.

Telemetry evidence **shall include**, at a minimum:

- SAPP simulation telemetry generated during pre-training and pre-deployment evaluation;
- runtime-parity control telemetry captured during live system operation, where applicable;
- human-in-the-loop interaction and intervention logs; and
- policy-as-code execution traces and enforcement outcomes.

Telemetry **shall be structured, machine-readable, timestamped, and attributable** to specific enforcement points and governance controls. Each telemetry domain **shall map one-to-one** with a

corresponding operational factor, ensuring that CSC scores reflect **observed system behavior rather than inferred intent**.

6.2 Factor-Level Scoring and Non-Compensatory Structure (Operational Factors 6.1–6.6)

CSC scoring is first calculated at the factor level, across six operational factors, each representing a distinct and essential governance domain:

- 6.1 Industry 4.0 & Security
- 6.2 Security & New Best Practice
- 6.3 Best Practice & Compliance
- 6.4 Compliance & Human-Centric Path
- 6.5 Human-Centric & Trust-Transparency
- 6.6 Trust-Transparency & Industry / Sector Alignment

Each factor shall be scored independently on a normalized 0–100 scale, based only on telemetry metrics relevant to that factor. No factor score is inferred from documentation or from performance in other domains.

The CSC star rating (★–★★★★★★) is not a direct representation of any single factor. Instead, it represents a composite assurance outcome derived from:

- the full set of factor-level scores,
- sector-specific weighting profiles, and
- mandatory qualification gates.

Individual factors do not map one-to-one to star levels. A higher star rating reflects stronger overall governance assurance, not the presence of specific features or controls.
CSC scoring is non-compensatory. Strong performance in one governance domain shall not offset weaknesses or failures in another. Where sufficient telemetry does not exist for a given factor, no score shall be assigned for that factor, and the maximum achievable CSC level shall be constrained or capped accordingly.
This structure ensures that:

- governance weaknesses cannot be hidden by high aggregate scores, and
- CSC ratings reflect balanced, evidence-based governance performance across all critical domains.

6.3 Aggregation Logic and Sector Weighting

CSC composite scoring **shall apply a weighted factor aggregation model** to reflect differing trust expectations across sectors. Default weightings **may be defined**, but **sector-specific profiles shall override defaults** where regulatory, safety, or societal risk differs materially.

Sector weighting **should account for**:

- severity and reversibility of potential harm;
- regulatory exposure and compliance obligations;

- public interest and critical infrastructure impact; and
- established industry governance norms.

All weighting models **shall be explicitly declared, versioned, and auditable**. Changes to sector profiles **shall not be applied retroactively** without re-issuance of the CSC certificate.

6.4 Gate Conditions and Star Qualification

CSC star levels **shall not be determined by aggregate score averages alone**. Mandatory qualification gates **shall be enforced** to ensure minimum acceptable governance performance.

Gate conditions **shall include**, at a minimum:

- minimum acceptable scores for designated critical factors;
- absence of unresolved critical control failures; and
- satisfaction of sector-specific trust requirements at higher certification levels.

Failure of any mandatory gate **shall cap the maximum achievable CSC star level**, regardless of composite score. This ensures that CSC certification reflects **minimum viable governance standards**, not statistical compensation across unrelated factors.

6.5 Confidence, Drift, and Continuous Re-Scoring

CSC certification **shall be continuous and lifecycle-aware**. Scores **shall be recalculated** as new telemetry is generated, enabling detection of system drift, control degradation, or changes in operational context.

Re-scoring mechanisms **shall account for**:

- telemetry completeness, freshness, and stability;
- consistency of control enforcement over time;
- effectiveness of human oversight mechanisms; and
- deviation from previously validated behavioral baselines.

Where telemetry confidence degrades or material drift is detected, CSC scores **shall be downgraded, suspended, or revoked** until corrective action is verified. This approach aligns CSC operation with ISO/IEC expectations for **continuous risk monitoring and treatment**, rather than point-in-time assessment.

Normative Statement (Section 6)

- CSC scoring **shall be derived exclusively from telemetry evidence**.

- Factor-level scoring **shall be non-compensatory**.
- Aggregation **shall be constrained by mandatory gates**.
- Certification **shall be continuously re-evaluated** in response to drift.
- If telemetry does not exist, the score shall not exist.

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7. Sector Anchoring and Industry Trust

AI systems operate within distinct sectoral contexts, each characterized by different risk profiles, regulatory obligations, and societal expectations. Sector anchoring ensures that governance, assurance, and trust measurement remain **context-aware**, **proportionate**, and **interpretable** across domains.

Within this framework, sector anchoring **shall not alter the core CSC methodology**. Instead, it refines how telemetry is weighted, interpreted, and governed, enabling comparability across sectors without imposing uniform risk assumptions.

7.1 Healthcare, Finance, and Public Sector Profiles

Sector profiles align governance mechanisms with **real-world impact, regulatory exposure, and domain-specific risk tolerance**. Each sector profile **shall define** how CSC scoring is contextualized, without modifying the underlying factor structure.

Sector-Specific Governance Characteristics

- **Healthcare**
  Governance **shall prioritize patient safety, clinical accountability, and explainability**. Telemetry emphasis should include human-in-the-loop intervention, decision traceability, and outcome validation. Risk tolerance is low, and assurance degradation **shall trigger immediate review**.
- **Finance**
  Governance **shall emphasize security, fraud prevention, auditability, and systemic risk control**. Telemetry should capture anomaly detection, control latency, and delegated decision accountability. Certification levels **shall reflect exposure to market and consumer harm**.
- **Public Sector**
  Governance **shall prioritize transparency, procedural fairness, contestability, and public accountability**. Telemetry emphasis should include decision explainability, appeal handling, and policy adherence across jurisdictions.

Customized Industry / Sector JSON Record Keeping

Sector anchoring **shall be operationalized through machine-readable sector profile records**, enabling automated interpretation and audit.

Sector JSON records **should include**, at a minimum:

- sector identifier and jurisdictional scope;
- sector-specific weighting overrides for CSC factors;
- mandatory gate conditions;
- telemetry emphasis indicators (e.g., human-centric, security-critical);
- applicable regulatory references; and
- versioning and effective date metadata.

These records **shall be versioned, auditable, and referenced** during CSC computation, ensuring that sector context is explicit rather than implicit.

7.2 Fair Competition, Credentialing, and Ecosystem Integrity

CSC supports fair competition by making governance performance **verifiable, comparable, and evidence-based**, rather than reliant on claims, branding, or market dominance.

Within this framework:

- trust **shall be earned through behavior**, not asserted through reputation;
- CSC scores **shall be comparable across organizations operating in the same sector profile**; and
- assurance **shall be portable**, subject to scope and evidence validity.

First Movers, Exemplars, and Expertise Sharing

The framework recognizes the role of **first movers and exemplars** in establishing early trust norms. Organizations demonstrating robust governance practices **may serve as reference implementations**, contributing patterns, tooling, and lessons learned to the broader ecosystem.

Expertise sharing **should be encouraged** through:

- publication of anonymized governance metrics;
- contribution of reusable policy templates; and
- shared learning from incident response and remediation.

Such contributions **shall not confer preferential certification status**, but they **may strengthen trust as a shared infrastructure** by accelerating collective maturity.

7.3 Trust as a Shared Infrastructure

Trust, within this framework, is not a proprietary asset but an **interoperable infrastructure layer** that spans organizations, platforms, and jurisdictions.

Trust as shared infrastructure implies that:

- governance signals **shall be machine-readable and interoperable**;
- assurance **shall persist across organizational boundaries**, subject to scope;
- trust **shall degrade consistently** when evidence degrades, regardless of provider; and
- no single actor **shall unilaterally define trust standards**.

CSC, sector profiles, and telemetry-based assurance together form a **distributed trust fabric**, enabling systems to reason about governance quality in real time, across heterogeneous ecosystems.

This approach supports innovation while reducing systemic risk, ensuring that trust scales **with adoption**, rather than eroding under complexity.

Normative Summary (Section 7)

- Governance **shall be sector-aware but methodologically consistent**.
- Sector context **shall be expressed through explicit, machine-readable profiles**.
- Trust **shall be verifiable, comparable, and portable within defined scope**.
- Trust **shall function as shared infrastructure**, not proprietary signaling.

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8. Governance Implications and Systems Cross-Over Validation

This section describes how telemetry-driven governance, sector anchoring, and the Compliance Star Certificate (CSC) reshape the roles of regulators, organizations, and engineers. It further introduces **systems cross-over validation**, whereby governance assurance is strengthened through **multi-party verification and shared evidence infrastructure**, rather than isolated audits.

8.1 Regulators: From Audits to Runtime Evidence

Under traditional governance models, regulators primarily assess compliance through **periodic audits, documentation reviews, and post-incident investigations**. These approaches provide limited visibility into ongoing system behavior and are poorly suited to adaptive, continuously evolving AI systems.

Within this framework, regulators **shall transition** from audit-centric oversight to **runtime evidence-based supervision**.

Specifically, regulators **shall be able to**:

- access continuous, machine-readable evidence derived from SAPP simulations and runtime telemetry;
- interpret CSC outputs as **current risk posture**, not historical approval;
- observe governance performance across system updates, scaling events, and cross-organizational deployment; and
- evaluate compliance drift as it occurs, rather than after harm has materialized.

Cross-Modal and Cross-Sector Visibility

Runtime evidence **may be exposed** to regulators through controlled, role-appropriate interfaces that support:

- cross-modal visibility (text, decision traces, control events, human review actions);
- longitudinal views of organizational maturity and scaling behavior; and
- sector-specific risk interpretation using declared sector profiles.

This approach enables regulators to **support development and scaling**, not merely enforce constraints, by identifying early risk signals and governance gaps across industries and organizational stages.

8.2 Organizations: Operating Trust at Scale

For organizations, trust is no longer a static attribute or reputational claim. Within this framework, trust **shall be managed as an operational metric**, comparable to reliability, safety, or security.

Organizations **shall**:

- treat CSC scores and factor-level outputs as live governance indicators;
- incorporate trust metrics into deployment, scaling, and procurement decisions; and
- respond to trust degradation as a material operational event.

Cross-Fusion Prop Validation and Industry Collaboration

To support scalable governance, organizations within the same sector **may participate in cross-fusion validation mechanisms**, whereby governance practices are compared, validated, and improved collaboratively.

Such collaboration **may include**:

- shared governance baselines within sector profiles;
- anonymized comparison of CSC factor trends;
- collaborative interpretation of regulatory expectations; and
- exchange of validated policy templates and control patterns.

Blockchain-Based Validation Infrastructure

Blockchain or distributed-ledger technologies **may be used** as a neutral validation substrate to:

- record CSC outputs, sector profiles, and evidence hashes;
- support tamper-evident governance records; and
- enable cross-organizational verification without central ownership.

In this model, organizations **may hold a defined number of validation nodes** (e.g., three), enabling peer verification while preserving organizational autonomy. Such mechanisms **shall augment**, not replace, regulatory oversight.

8.3 Engineers: Designing for Verifiability

For engineers, this framework shifts the design objective from *compliance after the fact* to **verifiability by construction**.

Systems **shall be designed** so that:

- governance controls are executable and observable;
- enforcement points are instrumented by default;
- telemetry is generated as a first-class output; and
- assurance does not depend on manual reconstruction.

Shared Validation Templates and Evidence Exchange

Engineers **may leverage shared, blockchain-anchored governance templates**, including:

- policy-as-code modules;
- telemetry schemas;
- CSC factor mappings; and
- validation test suites.

These templates **may be exchanged on a fixed cadence** (e.g., weekly), enabling convergence of governance practices across organizations without enforcing uniform system design.

Distributed Validator Model

To balance independence and collaboration:

- regulators **may operate a defined set of validators** (e.g., three);
- organizations **may operate a corresponding set of validators**; and
- governance evidence **shall converge on a fixed review interval** (e.g., one week).

This convergence process **shall support expertise transfer**, peer learning, and alignment of expectations. Each organization **shall retain accountability** through a designated governance and compliance signatory, ensuring that collaboration does not dilute responsibility.

Normative Summary (Section 8)

- Regulators **shall rely on runtime evidence**, not static audits alone.
- Organizations **shall manage trust as an operational metric**.
- Engineers **shall design systems to be verifiable by construction**.
- Cross-organizational validation **may be supported through shared, tamper-evident infrastructure**.
- Collaboration **shall strengthen governance without obscuring accountability**.

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9. Conclusion

This document has presented a framework for governing AI systems through **operational, evidence-based mechanisms** rather than static compliance artifacts. By integrating Simulation AI-Powered Policies (SAPP), the Access Policy-on-Demand (PoD) engine, and the Compliance Star Certificate (CSC), it establishes a foundation for **continuous assurance, proportional accountability, and human-centric oversight** across the AI lifecycle.

The framework is intentionally designed to remain **architecture-agnostic**, **sector-aware**, and **forward-compatible**, enabling adoption across diverse AI systems and organizational contexts.

9.1 Engineering Trust Before Impact

Trust in AI systems **shall not be retrofitted after deployment**, nor inferred from documentation or intent. It **shall be engineered before impact occurs**, and maintained through continuous observation, validation, and intervention.

By shifting governance upstream—into simulation, runtime enforcement, and telemetry-driven assurance—this framework enables:

- early identification of latent risk,
- timely corrective action as systems evolve, and
- sustained accountability across organizational and technical boundaries.

Engineering trust before impact reduces reliance on reactive enforcement and supports safer innovation without suppressing experimentation.

9.2 The Future of AI Governance Infrastructure

As AI systems become more adaptive, interconnected, and foundational to societal infrastructure, governance mechanisms **are expected to increasingly resemble** runtime safety and control systems, rather than periodic audit regimes.

Future AI governance infrastructure **shall**:

- rely on continuous evidence rather than point-in-time assessment;
- support interoperability of governance signals across organizations and sectors;
- enable conformance to be evaluated dynamically, not declared statically; and
- evolve through shared practice, validated outcomes, and cross-sector collaboration.

Within this context, conformance, interoperability, and standards evolution are **not endpoints**, but ongoing processes. Standards leadership will emerge from **demonstrated, repeatable governance practices**, reinforced through collaboration and transparent evidence exchange.

This document contributes a reference framework toward that evolution, one that invites scrutiny, refinement, and adoption through use, rather than authority - supporting the development of AI systems that are not only capable, but **trustworthy by design and accountable by operation**.

Final Normative Statement

- Trust **shall be engineered, observed, and maintained continuously**.
- Conformance **shall be evidence-based and revocable**.
- Interoperability **shall enable shared understanding without centralized control**.
- Standards **shall evolve from validated practice, not assertion**.

Conformance and Interoperability Clause

This clause defines how conformance to the framework specified in this document **shall be interpreted**, and how interoperability and cross-organizational collaboration **reinforce best practice and contribute to the emergence of leading governance standards**.

C.1 Conformance

An organization, system, or governance implementation **may claim conformance** to this framework if, and only if, the following conditions are met:

- Governance controls are implemented as **executable mechanisms** (e.g., policy-as-code) rather than solely as documented procedures.
- **Simulation AI-Powered Policies (SAPP)** are used to evaluate risk and control effectiveness prior to deployment or material system change.
- **Runtime telemetry** is generated, retained, and made available as evidence for governance assessment.
- **Compliance Star Certificate (CSC)** scoring is derived exclusively from telemetry and is subject to continuous re-evaluation.
- Human oversight mechanisms are integrated and auditable for high-impact governance decisions.

Conformance **shall not be claimed** on the basis of documentation, attestations, or declarations alone. Where required evidence is absent or degraded, conformance **shall be considered partial or invalid**.

C.2 Interoperability

Interoperability within this framework refers to the ability of governance mechanisms, evidence, and assurance signals to be **interpreted, compared, and validated across organizational and technical boundaries**.

To support interoperability:

- governance outputs **shall be machine-readable** and versioned;
- sector profiles **shall be explicitly declared** and referenced;
- CSC outputs **shall include scope, timestamp, and validity context**; and
- telemetry schemas **should align** with shared or published templates where available.

Interoperability **does not require uniform system architecture**. It requires **consistent interpretation of governance evidence**, enabling independent assessment by regulators, partners, and oversight bodies.

C.3 Cross-Organizational Collaboration and Best Practice Reinforcement

This framework recognizes that governance maturity is strengthened through **collaborative learning and comparative validation**, rather than isolated compliance efforts.

Accordingly:

- organizations **may participate** in cross-organizational governance validation within shared sector profiles;
- anonymized comparison of CSC factor trends **may be used** to identify systemic governance gaps or emerging risks;
- shared interpretation of regulatory expectations **should be encouraged**; and
- reusable policy patterns, telemetry schemas, and control templates **may be exchanged** to accelerate collective maturity.

Such collaboration **shall reinforce best practice**, without transferring accountability or creating implicit endorsement.

C.4 Emergent Standards and Leading Practice

This document **does not claim to define a final or exclusive standard**. Instead, it specifies a **reference framework** through which leading practices in AI governance may be operationalized, tested, and evolved.

Through consistent application, evidence exchange, and cross-sector collaboration:

- common governance patterns **may emerge** as de facto best practice;
- interoperability constraints **may inform future standardization efforts**; and
- regulators, industry, and researchers **may converge** on shared expectations grounded in runtime evidence.

In this way, the framework supports the **co-development of leading standards**, while remaining adaptable to technological and regulatory change.

Normative Summary (Conformance and Interoperability)

- Conformance **shall be evidence-based and continuously evaluated**.
- Interoperability **shall enable independent interpretation of governance evidence**.
- Cross-organizational collaboration **shall reinforce best practice without diluting accountability**.
- Standards leadership **shall emerge from validated practice, not assertion**.

References

ISO/IEC 42001 — Artificial Intelligence Management Systems

ISO/IEC 23894 — Artificial Intelligence Risk Management

ISO/IEC 22989 — Artificial Intelligence Concepts and Terminology

A Web of sharing trust responsibility – Policy of Developments – Access PoD vol.1

Appendix A — Methods

(Normative unless otherwise stated)

This appendix specifies the **formal method by which Simulation AI-Powered Policies (SAPP) telemetry is transformed into Compliance Star Certificate (CSC) scores** through metric normalization, aggregation, and evidence verification. It defines procedures and structures required for **evidence-based, continuous assurance**, and does not introduce additional governance principles beyond those stated in the main body of this document.

A.1 Purpose of the CSC Scoring Rubric

The Compliance Star Certificate (CSC) is a **runtime-evidenced trust assurance mechanism**, not a declarative or document-based compliance label.

The CSC scoring rubric defines how CSC scores **shall be computed directly from telemetry evidence**, ensuring that:

- certification reflects **observed system behavior**, not stated intent;
- scores are **continuously updateable** as telemetry changes;
- trust is **earned, measurable, and revocable**; and

- governance aligns with **ISO-style risk management**, rather than static assessment.

CSC scores **shall not be assigned manually**. All scores **shall be computed** using the procedures defined in this appendix.

A.2 CSC Scoring Architecture (High Level)

Inputs

CSC computation **shall consume** the following telemetry inputs, where applicable:

- SAPP simulation telemetry generated during pre-training and pre-deployment evaluation;
- runtime-parity control telemetry captured during live system operation;
- human-in-the-loop interaction and intervention logs; and
- policy-as-code execution traces and enforcement outcomes.

Processing

Telemetry inputs **shall be processed** through the following stages:

1. Metric normalization
2. Risk-weighted aggregation
3. Factor-level scoring
4. Star-level qualification gate evaluation

Outputs

CSC computation **shall produce**:

- a CSC Star Rating (★–★★★★★★);
- factor-level subscores corresponding to operational factors 6.1–6.6;
- an ISO-aligned risk tier classification; and
- a machine-readable evidence bundle supporting audit and verification.

A.3 Telemetry Domains

CSC scoring draws from **six telemetry domains**, each mapped **one-to-one** with a corresponding operational factor.

Each telemetry domain **shall generate**:

- quantitative metrics,
- qualitative signals, and
- event traces.

No factor score **shall be computed** where the corresponding telemetry domain is incomplete or unavailable.

A.4 Scoring Scale

Each operational factor **shall be scored on a normalized 0–100 scale**, derived from weighted telemetry metrics associated with that factor.

Scores **shall be comparable across systems** only within the same declared sector profile and scope.

A.5 Factor-Level Scoring Rubric

A.5.1 Factor 6.1 — Industry 4.0 & Security (Trust)

Telemetry sources include:

- abuse and misuse simulation outcomes;
- anomaly detection accuracy;
- incident response latency; and
- kill-switch and rollback activation success.

Core metrics include:

- percentage of simulated attacks detected;
- mean time to containment (MTTC);
- false positive and false negative rates; and
- policy enforcement success rate.

Scoring logic:

- <60: reactive security, weak containment
- 60–79: active controls, inconsistent outcomes
- 80–89: predictive detection, fast response
- ≥90: proactive, low-latency, high-fidelity defense

A.5.2 Factor 6.2 — Security & New Best Practice (Trust)

Telemetry sources include:

- control adaptation frequency;
- red-team and adversarial regression results;
- supply-chain integrity checks; and
- data-handling observability.

Scoring logic:

- <60: static security posture
- 60–79: periodic updates, limited learning
- 80–89: continuous hardening
- ≥90: self-improving security posture

A.5.3 Factor 6.3 — Best Practice & Compliance (Transparency)

Telemetry sources include:

- policy adoption ledgers;
- change simulation outcomes;
- deployment topology mapping; and
- drift detection logs.

Scoring logic:

- <60: opaque change management
- 60–79: documented but delayed adaptation
- 80–89: near-real-time transparency
- ≥90: fully observable, traceable lifecycle

A.5.4 Factor 6.4 — Compliance & Human-Centric Path (Transparency)

Telemetry sources include:

- human-in-the-loop interventions;
- escalation and override logs;
- risk classification outcomes; and
- audit evidence completeness.

Scoring logic:

- <60: nominal or symbolic human oversight
- 60–79: oversight present but inconsistent
- 80–89: structured, auditable governance
- ≥90: human oversight embedded and effective

A.5.5 Factor 6.5 — Human-Centric & Trust-Transparency (Openness)

Telemetry sources include:

- explainability responses;
- user impact reports;
- complaint and redress workflows; and
- outcome validation signals.

Scoring logic:

- <60: minimal user-facing transparency
- 60–79: reactive explanations
- 80–89: clear, timely trust signals
- ≥90: outcome-driven transparency

A.5.6 Factor 6.6 — Trust-Transparency & Industry / Sector (Openness)

Telemetry sources include:

- sector-specific policy simulations;
- fair-competition and anti-spoofing checks;
- credential verification outcomes; and
- ecosystem trust signals.

Scoring logic:

- <60: generic trust model
- 60–79: partial sector alignment
- 80–89: sector-calibrated governance
- ≥90: sector-validated trust posture

A.6 Aggregation Logic

A.6.1 Weighted Factor Model

CSC composite scoring **shall apply** a weighted factor model:

CSC Composite Score = Σ (Factor Score × Weight)

Weighting **shall be determined** by the declared sector profile.

A.7 Star-Level Qualification Gates

CSC star ratings **shall not be derived from averages alone**. Mandatory gate conditions **shall be evaluated** prior to star assignment.

Failure of any mandatory gate **shall cap** the maximum achievable CSC level, regardless of aggregate score.

A.8 Risk Tier Mapping

CSC star ratings **shall map directly** to ISO-aligned risk tiers as defined in the main body of this document.

A.9 Evidence and Audit Artifacts

Each CSC issuance **shall include** a machine-readable evidence bundle containing:

- telemetry summaries per factor;
- policy execution traces;
- simulation and risk analysis outputs;
- human review and escalation records; and
- a cryptographically signed CSC artifact.

Evidence artifacts **shall be continuously regenerable** and independently verifiable.

A.10 Key Differentiator

The CSC scoring rubric **cannot be satisfied by documentation alone**.

If telemetry does not exist, the score shall not exist.

This property makes CSC:

- hard to game;
- easy to audit; and
- inherently aligned with dynamic AI systems.

Informative Note — Further Documentation (Non-Normative)

Additional **informative materials**, including **worked conformance claim examples**, **JSON schemas**, **sector profile definitions**, and **implementation guidance**, are maintained online.
Readers seeking:

- illustrative conformance claims,
- machine-readable governance artifacts, or
- extended technical references

are encouraged to consult the supporting documentation available via the project website.
(This note is informative and does not introduce additional requirements.)

Appendix B — Download Pack

The public download pack includes CSC JSON schemas, metric catalogs, sector profiles, example certificates, and Access PoD implementation guidance.

The GitHub release link represents the canonical public review artifact set for Version A.

Access PoD Artifact Pack – v1a (Public Review Draft).
GitHub Release.
https://github.com/Access-PoD/access-pod-artifacts/releases/tag/v1a

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Authorship and AI-Assisted Tools Disclosure (Informative)

This document was authored by the named author. AI-assisted tools were used in a limited capacity to support language refinement, structural consistency, and standards-aligned phrasing.

Core concepts, governance architecture, scoring methodology, and system design originate from the author. AI tools did not originate any novel frameworks, models, or governance mechanisms described in this document.